

Facility: SALEM 1 & 2 Scenario No.: ESG-4 Op-Test No.: 13-01 NRC

Examiners: _____ Operators: _____

Initial Conditions: 4% power, BOL. Unit is returning from forced outage to repair all 4 RCP diffuser vanes bolting issue. 24 SW pump is C/T for strainer repair.

Turnover: Raise power to 18% @10% per hour.

Event No.	Malf. No.	Event Type*	Event Description
1		R ALL	Raise power
2	SW0216A	C ALL	#2 SW bay leak (TS)
3	RC0002	C ALL	RCS leak (TS)
4	CV0208A CV0208B	C CRS/RO	Centrifugal charging pump fails during swap from PDP while responding to RCS leak. (TS)
5	RC0002	M CRS/RO	SBLOCA
6	EL0134	C ALL	LOOP coincident with SI initiation
7	O/R A701 B DI	C CRS/PO	2C SEC fails to initiate
8	RP318S1 RP318S2	C CRS/RO	High head ECCS fails to inject (one pump failed from Event 4 above, other pump doesn't start)
			CT's: #1 Establish High Head ECCS #2 Start minimum SW pumps

* (N)ormal, (R)eactivity, (I)nstrument, (C)omponent, (M)ajor

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 13-01 NRC ESG-4

SCENARIO NUMBER: 13-01 NRC ESG-4

EFFECTIVE DATE: See Approval Dates below

EXPECTED DURATION: 65 minutes

REVISION NUMBER: 00

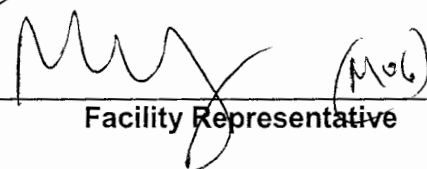
PROGRAM: ☐ L.O. REQUAL

☒ INITIAL LICENSE

☐ STA

☐ OTHER _____

Revision Summary
New Issue for 13-01 NRC Exam

PREPARED BY:	<u>G Gauding</u> Lead Regulatory Exam Author	<u>09-14-14</u> Date
APPROVED BY:	 Operations Training Manager	<u>10-23-14</u> Date
APPROVED BY:	 Facility Representative	<u>10-23-14</u> Date

I. OBJECTIVES

- A. Given the order or indications of a loss or malfunction of a safety related plant cooling water system, perform actions as the nuclear control operator to RESPOND to the loss or malfunction of the safety related plant cooling water system in accordance with S1/S2.OP-AB.SW-0001.
- B. Given indication of a loss or malfunction of a safety related plant cooling water system, DIRECT the response to the loss or malfunction in accordance with the approved station procedures.
- C. Given the indications of a reactor coolant system (RCS) malfunction or leak, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S1/S2.OP-AB.RC-0001.
- D. Given the indications of a reactor coolant system (RCS) malfunction or leak, DIRECT the response to the malfunction in accordance with S1/S2.OP-AB.RC-0001
- E. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.
- F. Given the order or indications of a reactor trip perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.
- G. Given indication of a safety injection DIRECT the response to the safety injection in accordance with the approved station procedures.
- H. Given the order or indications of a safety injection perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- I. Given a safety injection has occurred and equipment has failed to START equipment that has failed to automatically start in accordance with the approved station procedures.

II. MAJOR EVENTS

- A. Raise power
- B. #2 SW Bay leak
- C. RCS leak (25 gpm)
- D. Centrifugal charging pump failure
- E. SBLOCA
- F. LOOP coincident with SI initiation
- G. 2C SEC fails to initiate
- H. High Head ECCS fails to inject

III. SCENARIO SUMMARY

- A. The crew will take the turnover at 4% power, BOL, with direction to raise power to 18% @ 30% per hour. 24 SW pp is C/T for strainer repair. 21 SGFP is in service, 22 SGFP is latched at idle speed.
- B. After power ascension is in progress, a leak in #2 SW Bay occurs. The CRS will enter S2.OP-AB.SW-0003, Service Water Bay leak, cross connect nuclear headers and isolate the affected SW bay. The CRS will identify Tech Spec 3.0.3 is applicable based on SW pump operability.
- C. After the SW Bay leak has been addressed, a 25 gpm RCS leak will occur. The CRS will enter S2.OP-AB.RC-0001, Reactor Coolant System Leak. When swapping to a centrifugal charging pump, the pump selected will trip after starting, and the other centrifugal charging pump will be placed in service. The crew will quantify the RCS leakrate, and the CRS will identify Tech Specs.
- D. Once the RCS leak has been addressed, the leak will worsen into a SBLOCA. The CRS will direct a Rx trip and safety injection. If attempted first, Train B SI initiation will not work, and Train A will be used to initiate SI.
- E. When SI is initiated, a loss of off-site power occurs. 2A and 2B SECs will load in MODE III (SI plus Blackout). 2C SEC fails to initiate. Operators will manually shut 2C EDG output breaker to energize 2C vital bus. The crew will start at least 1 SW pump on 2C bus. The crew will start one charging pump to establish high head ECCS injection.
- F. The crew will perform Rx trip stabilization actions in TRIP-1, then transition to LOCA-1 to reset SI and re-establish control of plant equipment. The scenario will terminate when the TRIP-3 transition point is reached.

IV. INITIAL CONDITIONS

____ Pre-snapped IC-234

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

<i>Initial</i>	<i>Description</i>
____ 1	RCPs (SELF CHECK)
____ 2	RTBs (SELF CHECK)
____ 3	MS167s (SELF CHECK)
____ 4	500 KV SWYD (SELF CHECK)
____ 5	SGFP Trip (SELF CHECK)
____ 6	23 CV PP (SELF CHECK)
____ 7	24 SW pump C/T
____ 8	S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load, filled out up to step 5.3.16.
____ 9	S2.OP-SO.MS-0002, Steam Dump System Operation, with P&L 2.2 signed off.
____ 10	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
	3	EVENT ACTION: kb417pbj // 21 CHG PUMP-START COMMAND: DMF CV0208B PURPOSE: <update as needed>
	5	EVENT ACTION: kb419pbj // 22 CHG PUMP-START COMMAND: DMF CV0208A PURPOSE: <update as needed>
	7	EVENT ACTION: ka501doa // TRAIN 'A' - SI OPERATE KEYSWITCH COMMAND: PURPOSE: <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
___ 01	RC0002 RCS LEAK INTO CONTAINMENT (equiv to 0-4 inches)	N/A	N/A	N/A	RT-3	25
___ 02	CV0208A 21 CHARGING PUMP TRIP	00:00:03	N/A	N/A	ET-3	
___ 03	CV0208B 22 CHARGING PUMP TRIP	00:00:03	N/A	N/A	ET-5	
___ 04	EL0134 LOSS OF ALL 500KV OFF-SITE POWE	N/A	N/A	N/A	ET-7	
___ 05	RP318S1 21 CHR G PMP FAILS TO STRT ON SEC	N/A	N/A	N/A	N/A	
___ 06	RP318S2 22 CHR G PMP FAILS TO STRT ON SEC	N/A	N/A	N/A	N/A	
___ 07	SW0216A 21 SW HDR LEAK IN SW STRUCTURE	N/A	0	00:03:00	RT-1	8000

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
___ 01	SW35D 24 SW PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF
___ 02	SW36D 24 SW PUMP RACK OUT	N/A	N/A	N/A	N/A	TAGGED
___ 03	SW23D 21 SW PUMP BKR CONTROL POWER	N/A	N/A	N/A	RT-10	OFF
___ 04	SW27D 22 SW PUMP BKR CONTROL POWER	00:00:10	N/A	N/A	RT-10	OFF
___ 05	SW32D 23 SW PUMP BKR CONTROL POWER	00:00:20	N/A	N/A	RT-10	OFF

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
___ 01	A701 B DI TRAIN 'B' - SI OPERATE KEYSWITCH	N/A	N/A	N/A	N/A	OFF
___ 02	C206 A DI 2C SFGD EMER OPR-EMERG LOADING RESET	N/A	N/A	N/A	N/A	ON
___ 03	AH03 OVDI 22 SI PUMP START	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:

	Description
___ 1.	None

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Power ascension Note: There is no limit on power ascension rate when < 50% power. Note: S2.OP-SO.MS-0002 allows steam dumps to be placed in Manual for adjustment. Manual allows for finer control. Continue to next event on direction from Lead Evaluator.			
	CRS briefs crew on power ascension.		
	PO slowly adjusts Main Steam dumps to raise reactor power using Attachment 3, by either adjusting the auto setpoint, or by placing steam dumps in manual and adjusting valve position.		
	RO slowly withdraws control rods.		
	Crew announces when Rx power is 5% and notes Mode 1 entry.		
	CRS contacts work control to update WCM to Mode 1.		
	PO ensures 21-24MS10's in auto set at 1015-1020 psig.		
	CRS continues power ascension.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
2. #2 SW Bay leak			
Simulator Operator: Insert RT-1 on direction from Lead Evaluator. MALF: SW0216B 22 SW HDR LEAK IN SW STRUCTURE Severity: 8,000 Ramp: 3 minutes			
Note: The first OHA will annunciate 1 minute after the leak is inserted.			
	RO announces unexpected SW system alarms.		
	CRS directs power ascension put on hold.		
	PO places Main Steam dumps in auto at current steam pressure if in manual.		
	RO monitors reactor.		
	PO either starts a SW pump based on lowering SW header pressure, or announces the auto start of 25 SW pump when it occurs, and the clearing of the low SW header pressures alarms.		
	PO reports indications of SW leak in #2 SW Bay.		
	CRS enters S2.OP-AB.SW-0003 SW Bay Leak per ARP direction or enters S2.OP-		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Role Play: 3 minutes after being contacted, report as NEO that there is a large leak in 2 bay, you can't identify where it is coming from due to the spray, and approx. 2.5 ft of water is in the bay now, and rising slowly. (Bay is considered disabled at 2 ft.	AB.SW-001, Loss of SW Header Pressure, then transitions to S2.OP-AB.SW-0003, and initiates CAS.		
	RO reports both SW Bays are in service and dispatches NEO to investigate 2 Bay alarms.		
	PO monitors Main Turbine / SGFP temps on SPDS and Plant Computer. (CAS Item 2.0)		
	Crew splits SW Bays by opening nuclear header cross over MOVs 21SW23 and 22SW23, and shutting SW bay cross connect valves 21SW17 and 22SW17, and reports the leak is in 2 Bay.		
	PO reports 25 and 26 SW pumps running, and stops 21-23 SW pumps.		
	Crew sends NEO to open control power bkrs for 21-23 SW pumps.		
Simulator Operator: 2 minutes after being directed to de-energize control power to 21-23 SW pumps, call as NEO and state: "This is			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
(NEOs name) calling for a First Check. I am in Unit 2 64' switchgear room, and the walls are yellow." After acknowledgment from control room, then INSERT RT-10: REMOTES: SW23D, SW27D, SW32D for 21, 22, and 23 SW Pump BKR Control Power with delays of 0, 10, and 20 seconds. Call control room after last remote is active and report control power breakers have been opened for 21, 22 and 23 SW pumps.			
Role Play: Once 2 SW Bay is isolated, NEO (if contacted) reports that the leak appears to have stopped, and level is slowly lowering in bay.	Crew isolates 2 SW Bay by closing 21SW22 and 21SW20.		
	PO reports the leak is isolated.		
	PO reports SW remains available to the EDGs.		
	PO reports SW is being supplied to the Turbine Building through 23SW20.		
	PO monitors Main Turbine parameters on SPDS and the Plant Computer. (CAS item 2.0)		
	PO initiates action to remove one CCHX from		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: CRS should contact work control to find status 24 SW pump, and when pump could be restored to service to avoid a unit shutdown due to Tech Spec 3.0.3. If asked, respond that the release for 24 SW pump is in the field. Proceed to next event after Tech Spec has been identified at Lead Evaluators direction. Note: Entry conditions are also met for S2.OP-AB.ZZ-002, Flooding, due to flooding in 2 Bay. Proceed to next event on direction from Lead Evaluator. 3. RCS leak Simulator Operator: Insert RT-3 on direction from Lead Evaluator. MALF: RC0002 RCS leak into containment Severity: 25	service. (CAS item 4.0)		
	PO initiates action to isolate SW to two CFCU's IAW CAS item 5.0.		
	CRS enters TSAS 3.0.3 due to having NO operable SW loops.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports that charging flow is rising and PZR level is lowering slowly.		
	Crew reports reading on 2R11A containment radiation monitor is rising.		
	RO reports unexpected OHA C-2 CNTMT SUMP PMP START.		
	CRS enters S2.OP-AB.RC-0001, Reactor Coolant System Leak.		
	CRS directs implementation of Att. 1 CAS.		
	RO reports RCS temperature >350°F.		
	RO reports unit is in MODE 1.		
	RO reports PZR level is lowering with maximum PDP flow.		
	RO transfers to a centrifugal charging pump by performing step 3.14 as follows:		
	RO ensures charging master flow controller in auto.		
	RO shuts 2CV55.		
Simulator Operator: Ensure ET-3 OR ET-5 is true when 21 or 22 charging pump is started. This will trip the charging pump which was started, and delete the trip of the			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
remaining charging pump allowing its start.			
4. Centrifugal charging pump trip			
	RO starts 21 or 22 charging pump, and reports it has tripped.		
	CRS directs RO to start remaining charging pump.		
	RO starts the remaining available centrifugal charging pump.		
Simulator Operator: After the second charging pump is started, reinsert the charging pump trip for the charging pump which tripped. If 21 charging pump is tripped, insert MALF CV0208A . If 22 charging pump is tripped, insert MALF CV0208B . This must be done prior to the SI initiation.			
	RO places 23 charging pump speed controller in manual.		
	While lowering 23 Charging Pump speed to minimum, RO adjusts 2CV55 to maintain desired flow.		
	RO stops 23 Charging Pump when at minimum flow		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Letdown will isolate if PZR level is allowed to lower to 17%.	RO adjusts 2CV55 to control Pressurizer level.		
	RO places 2CV55 in AUTO OR leaves in MANUAL until directed by the SM/CRS.		
	RO adjusts RCP Seal Injection flows to maintain 6-12 gpm per pump/		
	PO swaps to the 45 gpm letdown orifice.		
	RO estimates leak rate, and reports PZR level can be maintained stable or rising.		
	CRS initiates S2.OP-ST.RC-0008, Reactor Coolant System Water Inventory Balance.		
	CRS evaluates containment conditions and determines CFCU status.		
	CRS initiates actions to locate and isolate the leak IAW Attachment 2.		
	CRS enters TSAS 3.4.7.2.b. action b for the unidentified RCS leak, and 3.5.2.a action a for ECCS		
Note: Although the leak is within the makeup capacity of the CVCS system, a containment entry to locate and isolate should be ruled out due to the magnitude of the			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
leakage.			
Continue to next event on direction from Lead evaluator.			
5. SBLOCA			
Simulator Operator: On direction from Lead Evaluator, MODIFY MALF RC0002 from 25 to 2,000 with no ramp or delay.			
	RO reports indications of worsening RCS leak, and rapidly lowering PZR level.		
	CRS directs a Rx trip and Safety Injection.		
	RO trips the reactor, and confirms the Rx trip.		
	RO initiates SI on Train B and reports SI has not initiated.		
Note to Evaluators: Train B SSPS is closest to the Rx trip handles on the Control Console, and typically SI is initiated from Train B. This was not considered an event on Form ES-D-1 since it may not be performed by all operators, but may be counted as another component failure if performed.			
	RO initiates SI on Train A and reports SI has initiated on both trains.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: Ensure ET-7 is true when SI is initiated on Train A. This inserts the Loss of Off-Site Power.			
6. LOOP coincident with Safety Injection			
	Crew diagnoses loss of all off-site power when control room normal lighting goes out.		
7. 2C SEC fails to initiate	RO continues immediate actions: - Reports Main Turbine is tripped. - Reports 2A and 2B EDGs have started and are supplying respective vital buses. - Reports SI has been manually initiated.		
	CRS and RO verify immediate actions complete.		
	PO reports no conditions present that should keep 2C EDG from starting.		
	CRS directs PO to start 2C EDG and shut 2C EDG output breaker.		
	PO starts 2C EDG, selects bezel mimic for 2C EDG output breaker.		
	PO shuts 2C EDG output breaker.		
Note: With the 2C SEC not initiating, whatever equipment was			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
running on 2C bus before the LOOP will restart since no trip signals were sent to their breakers. Crew should open breakers for C bus equipment prior to closing EDG output breaker to prevent possibly stalling 2C EDG.			
CT#1 (CT-25) Start at least one SW pump prior to end of scenario. SAT _____ UNSAT _____			
8. High Head ECCS fails to inject			
	PO throttles AFW flow, and reports all SG NR levels are >9%.		
	PO reports SEC loading status for energized vital buses. If 22 charging pump was running before LOOP, then it will have restarted from C bus if the EDG breaker was shut.		
CT#2 (CT-7) Establish Flow from at least one High-head Charging Pump prior to transition out of TRIP-1. SAT _____ UNSAT _____	If 21 charging pump was running prior to the LOOP, the PO will block and reset 2B SEC, then start 21 charging pump.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	If 2C EDG breaker is shut, then crew will also need to evaluate SEC loading on 2C bus, since SEC will not have started any equipment, and only equipment which was previously running will be running now.		
	PO reports 21 and 22 AFW pumps running.		
	Crew recognizes when containment pressure reaches 4 psig and uses adverse containment values.		
	RO shuts charging pump mini flow valve 2CV139 and reports 2CV140 has no power when RCS pressure lowers below 1500 psig with BIT flow established. (CAS)		
	PO reports safeguards valve alignments.		
	RO reports 21/22CA330's shut and containment pressure has remained below 15 psig.		
	PO reports no high steam flow conditions.		
	PO reports 4KV vital bus status. If 2C EDG output breaker has not been shut, CRS directs power restored to 2C vital bus.		
	RO reports correct ventilation lineups.		
	RO reports CCW pump status. 21 and 22 should not be running since they were stripped		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: ECCS injection flow will be causing RCS temperature to lower. A MSLI is not required.	of 2A and 2B vital buses on SEC Mode III initiation. 23 CCW pump status will hinge on whether 2C EDG output breaker is shut.		
	RO reports ECCS injection status for current RCS pressure.		
	PO reports all SG NR levels are >9%. (15% adverse)		
	RO reports RCPs stopped, and RCS temperature being controlled.		
	RO reports reactor trip breakers are open		
	RO reports PZR PORVs are shut and Block Valves open.		
	RO reports RCPs are not in service.		
	PO reports no indication of faulted steam generators.		
	PO reports no indication of ruptured steam generators.		
	RO reports at least 2 channels in Table F in warning, alarm, or rising.		
	CRS transitions to LOCA-1, Loss of Reactor Coolant.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Terminate scenario after transition to LOCA-1 is made.			

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load
- F. NF-AP-440, PWR Fuel Conditioning
- G. S2.OP-AB.SW-0001, Loss of Service Water Header Pressure
- H. S2.OP-AB.SW-0003, Service Water Bay Leak
- I. S2.OP-AB.RC-0001, Reactor Coolant System Leak
- J. 2-EOP-TRIP-1, Reactor Trip or Safety Injection

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 4.1% RCS BORON: 1740 ppm MWe 0

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Core Burnup: 500 EFPH

Control Bank D is at 157 steps.

Xenon building in at 5 pcm / hr.

Reactivity plan for power ascension to 18.0% is to use S2.OP-SO.MS-0002, Steam Dump Operation, Section 5.4.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load, in progress and complete up to Step 5.3.16. All requirements for Mode 1 entry are satisfied.

ABNORMAL PLANT CONFIGURATIONS:

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

SECONDARY:

Main Steam Dumps are in MS Pressure Control-Auto set at 986 psig.

All MSIVs are open.

21 SGFP in service, 22 SGFP latched at idle speed. AFW secured.

Heating steam is aligned to Unit 1.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

24 SW pump C/T for strainer repair.

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
___ With Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct if applicable.
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3

CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant. The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 13-01 NRC-ESG-4 **REVIEWER:** P Williams

Initials	Qualitative Attributes
PW	1. The scenario has clearly stated objectives in the scenario.
PW	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
PW	3. The scenario consists mostly of related events.
PW	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
PW	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
PW	6. The events are valid with regard to physics and thermodynamics.
PW	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
PW	8. The simulator modeling is not altered.
PW	9. All crew competencies can be evaluated.
PW	10. The scenario has been validated.
PW	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
PW	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

DO NOT WRITE IN THESE SPACES

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)
GG	7	Total malfunctions inserted: 5-8
GG	3	Malfunctions that occur after EOP entry: 1-2
GG	2	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	1	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	Crew Critical Tasks: 2-3

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

13-01 NRC ESG-4

CT#1 (CT-25) Start at least one SW pump prior to end of scenario.

Bases: Failure to manually start the SW pump under the postulated plant conditions means that the EDG's are running without SW cooling. Running the EDGs without SW cooling leads to a high-temperature condition that can result in EDG failure due to damage caused by engine overheating. Under the postulated plant conditions, all running EDGs are affected by the loss of all SW cooling. Thus, failure to perform the critical task constitutes misoperation or incorrect crew performance in which the crew does not prevent "degraded... emergency power capacity."

CT#2 (CT-7) Establish Flow from At Least One High-head SI Pump prior to transition out of TRIP-1.

Bases: Failure to manually start at least one high-head SI pump under the postulated conditions constitutes misoperation or incorrect crew performance in which the crew does not prevent "degraded emergency core cooling system (ECCS) ...capacity." In this case, at least one high-head SI pump can be manually started from the control room. Therefore, failure to manually start a high-head SI pump also represents a failure by the crew to "demonstrate the following abilities:

- Effectively direct or manipulate engineered safety feature (ESF) controls that would prevent a significant reduction of safety margin beyond that irreparably introduced by the scenario
- Recognize a failure or an incorrect automatic actuation of an ESF system or component"

EVENTS LEADING TO CORE DAMAGE

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
N	Steam Generator Tube Rupture	N	Loss of CCW
Y	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
Y	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	N	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	N	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.